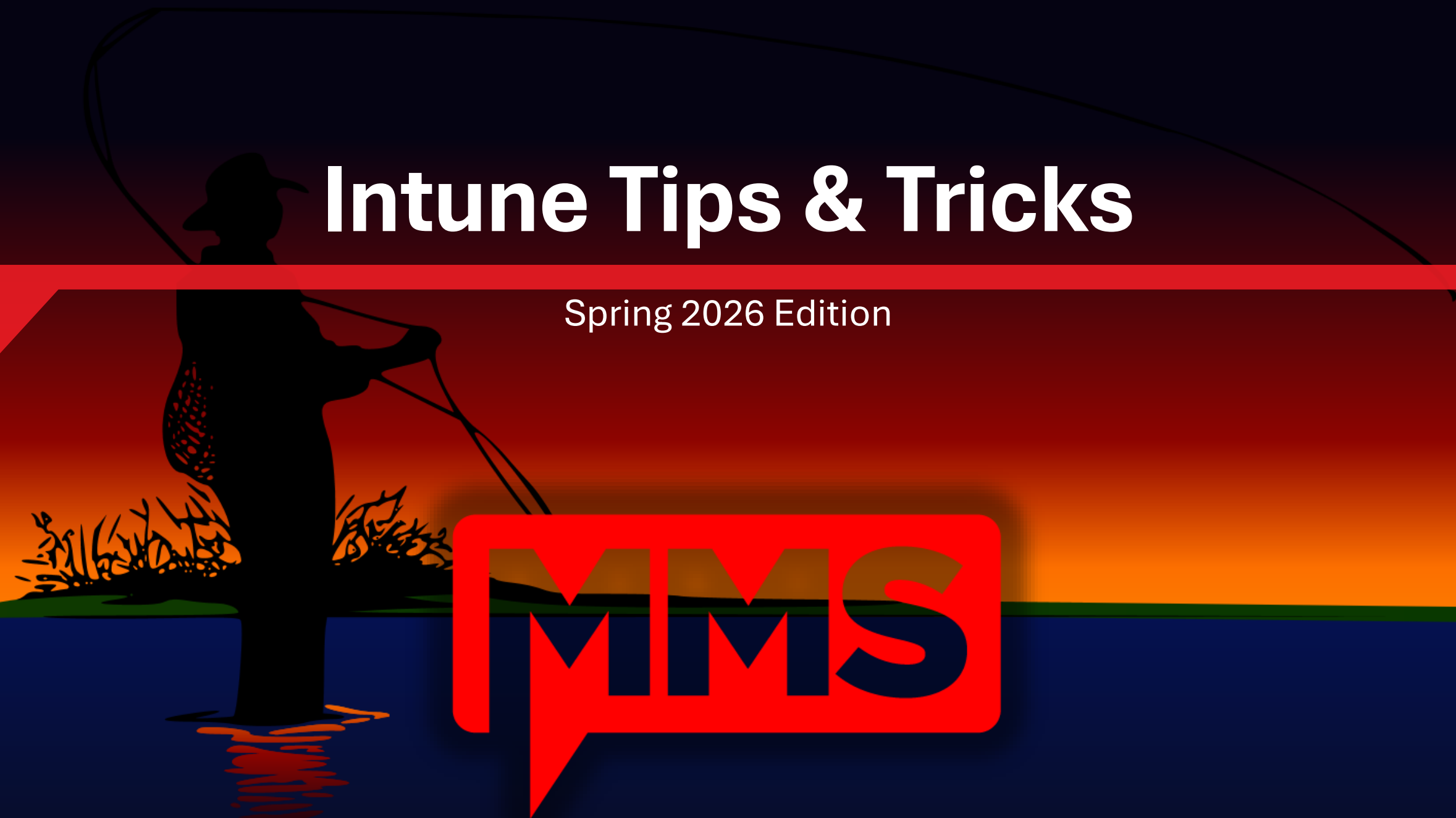


A silhouette of a fisherman wearing a hat and waders, standing in water and casting a long net. The background is a gradient from dark blue at the top to orange and red in the middle, with a dark blue water surface at the bottom. The fisherman's net is arched high in the air.

Intune Tips & Tricks

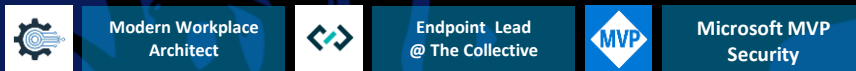
Spring 2026 Edition



Speakers



Tim De Keukelaere



 Tim.De.Keukelaere@it-essence.be



<https://linktr.ee/timdk>



Peter Daalmans

Modern Workplace Architect, MVP, MCT

 peter@secminds.nl



<https://ref.ms/pdaalmans>

Can't see our slides? Can't hear? Need to repeat the question? Call us out!



BE AWARE



R
PR

E

V

I

E

W

Creating visibility



What's on the horizon ?

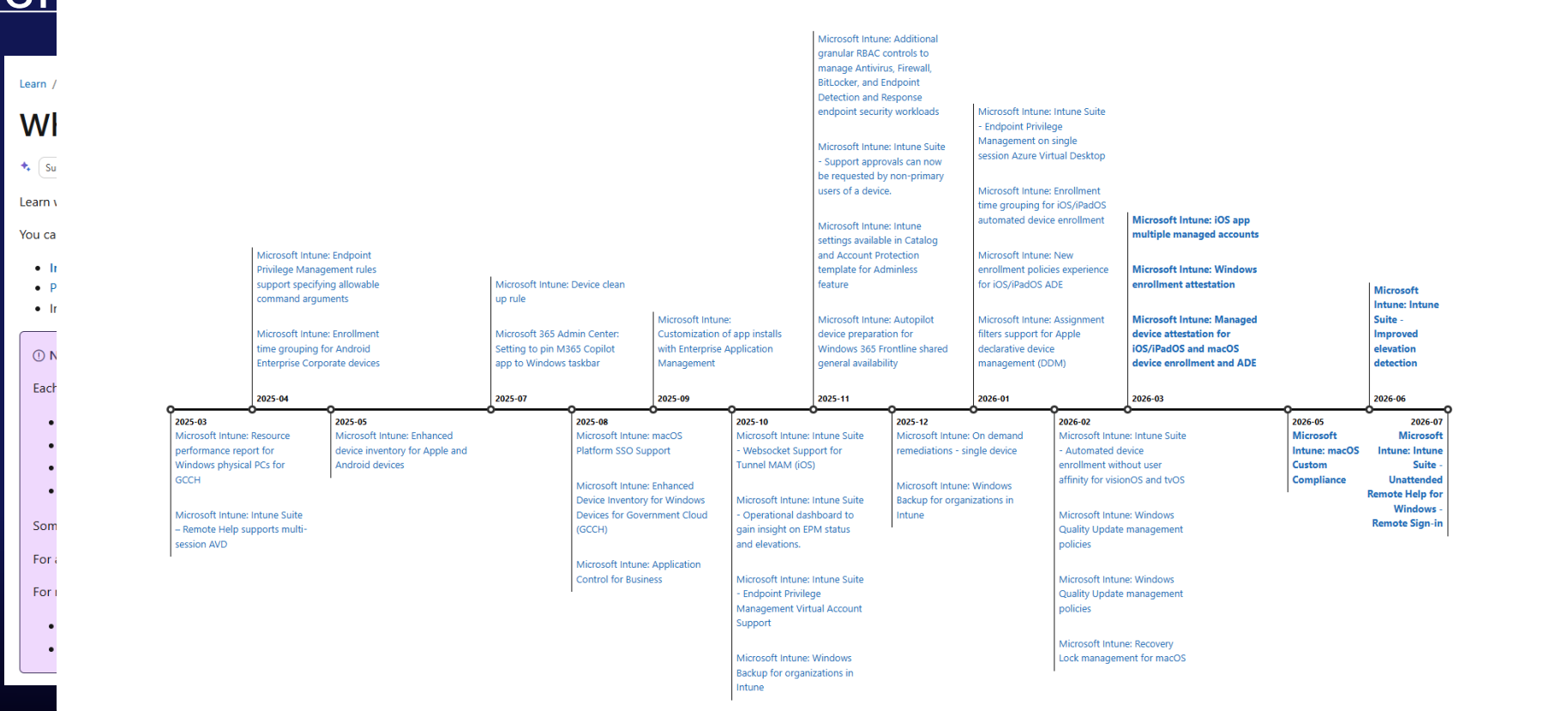
◆ What's on the horizon ?

● M365 Visual Roadmap

This site provides a visual representation of the Microsoft 365 Roadmap.

[View on GitHub](#)

If only there was a way to visualize this properly ...



<https://m365visualroadmap.net/roadmap/Intune.html>

Or put it all together?

One place to see:

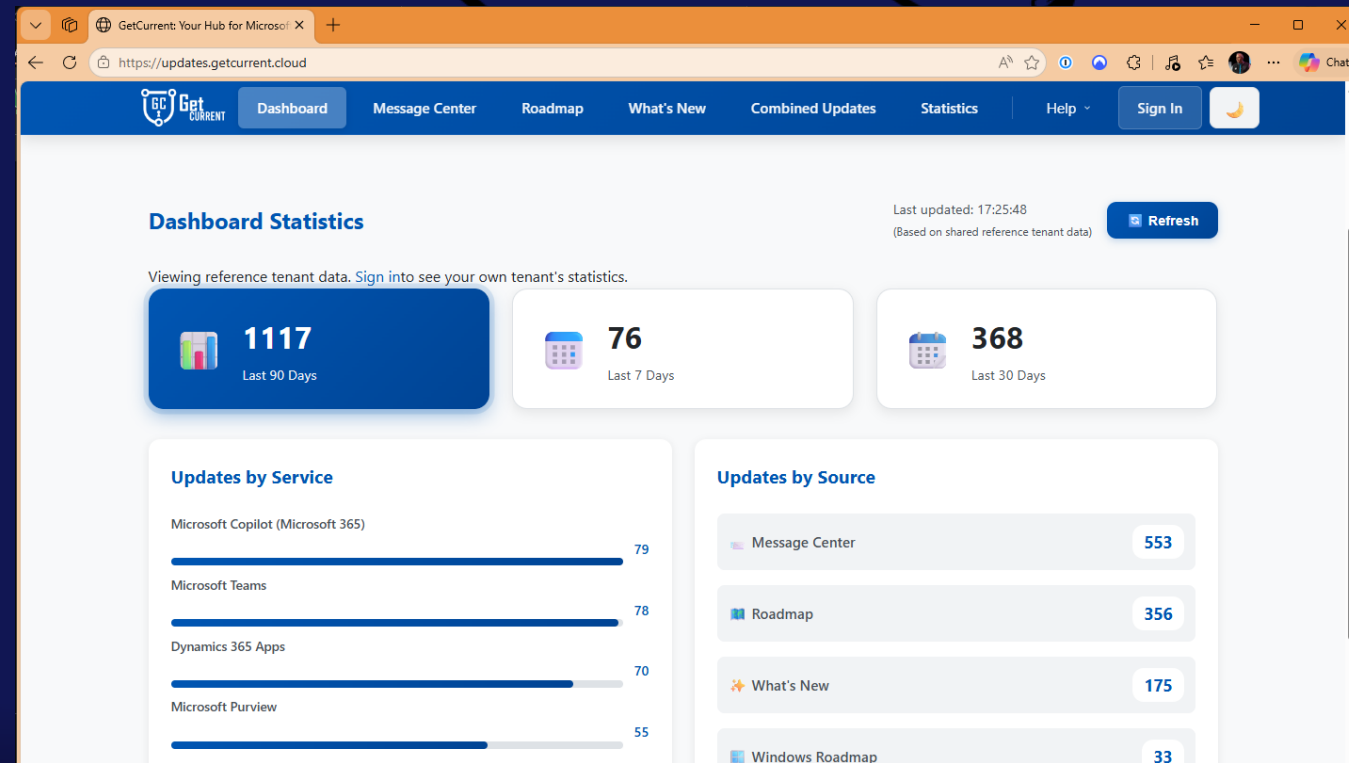
What's new on:

- Windows / Windows 365
- Intune
- Entra ID
- Defender
- Purview

The Roadmap of:

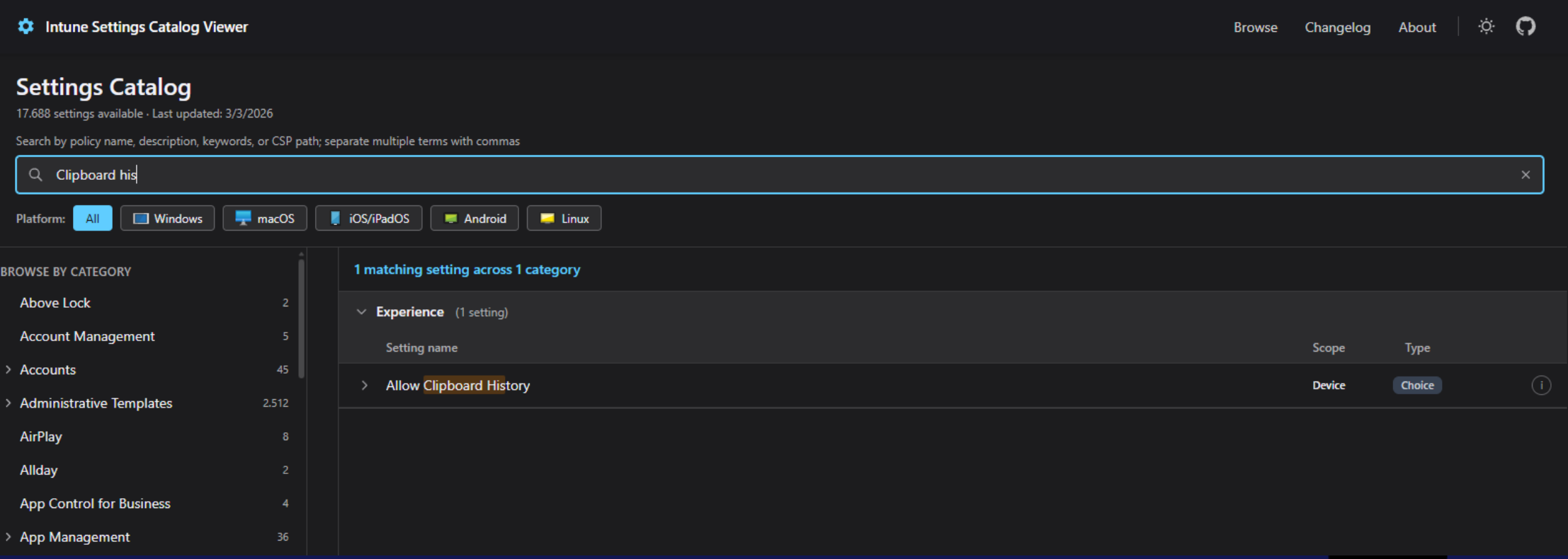
- Microsoft 365
- Windows

Message Center.



<https://updates.getcurrent.cloud>

Where's that setting?



The screenshot shows the 'Intune Settings Catalog Viewer' application. At the top, there's a header with a gear icon, the title 'Intune Settings Catalog Viewer', and links for 'Browse', 'Changelog', and 'About'. Below the header, the main section is titled 'Settings Catalog' with a subtitle '17,688 settings available · Last updated: 3/3/2026'. A search bar contains the text 'Clipboard his'. Below the search bar, there are platform filters: 'All' (selected), 'Windows', 'macOS', 'iOS/iPadOS', 'Android', and 'Linux'. On the left, a 'BROWSE BY CATEGORY' sidebar lists various categories with their respective counts. The main content area shows '1 matching setting across 1 category'. Under the 'Experience' category (1 setting), a table lists the setting 'Allow Clipboard History' with a scope of 'Device' and a type of 'Choice'.

Intune Settings Catalog Viewer

Browse Changelog About

Settings Catalog

17,688 settings available · Last updated: 3/3/2026

Search by policy name, description, keywords, or CSP path; separate multiple terms with commas

Clipboard his

Platform: All Windows macOS iOS/iPadOS Android Linux

BROWSE BY CATEGORY

- Above Lock 2
- Account Management 5
- > Accounts 45
- > Administrative Templates 2,512
- AirPlay 8
- Allday 2
- App Control for Business 4
- > App Management 36

1 matching setting across 1 category

Experience (1 setting)

Setting name	Scope	Type
> Allow Clipboard History	Device	Choice

<https://intunesettings.app>

DEMO

Creating visibility



Secure Boot certs





What is Secure Boot?

The Foundation of Boot-Time Trust

UEFI firmware security feature

Validates every piece of software in the boot chain before it runs.

Public Key Infrastructure (PKI)

Microsoft and OEM Certificate Authorities issue trusted digital certificates stored in firmware.

Signature check at every stage

Firmware drivers, bootloaders, and option ROMs must present valid signatures — unsigned or tampered code is blocked before Windows loads.

Hardware-anchored trust

Trust is rooted in firmware, not the OS — attackers can't bypass it from software.

Threats Secure Boot Prevents

Bootkits & Rootkits

Malware targeting bootloader or kernel at startup

Firmware Attacks

Tampering or replacing UEFI firmware components

Code Injection

Unauthorized code injected into the boot sequence

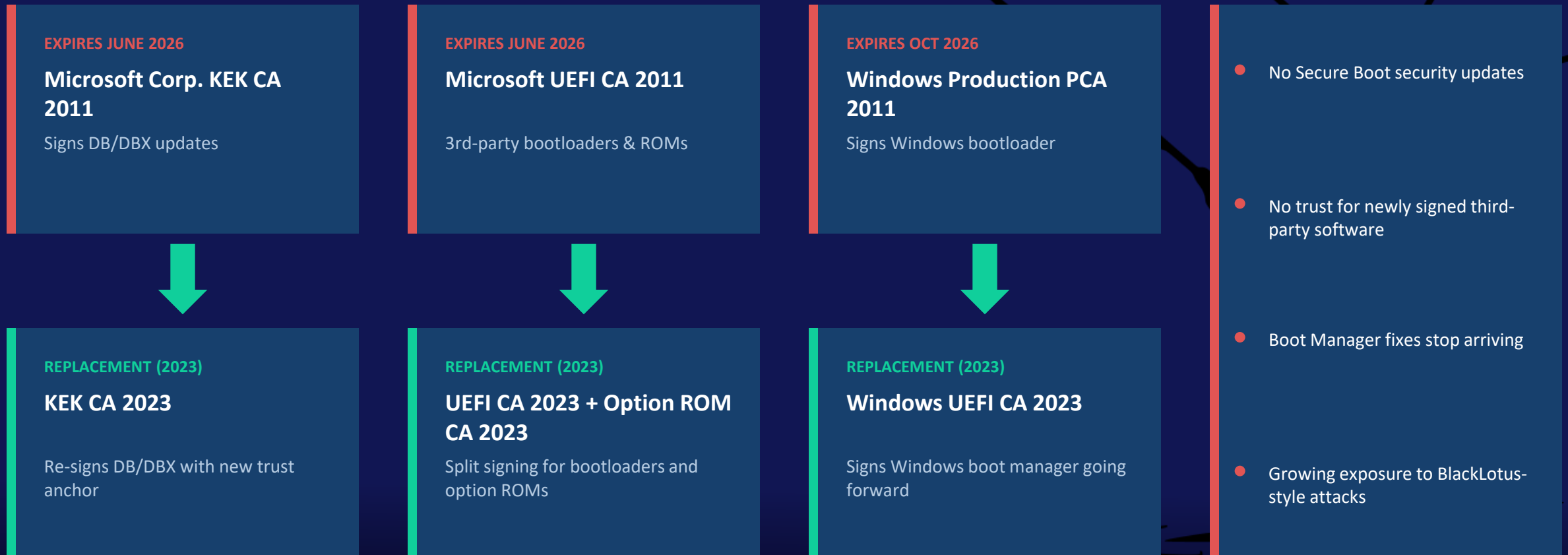
Unsigned Kernel Modules

Loading tampered or unverified kernel modules

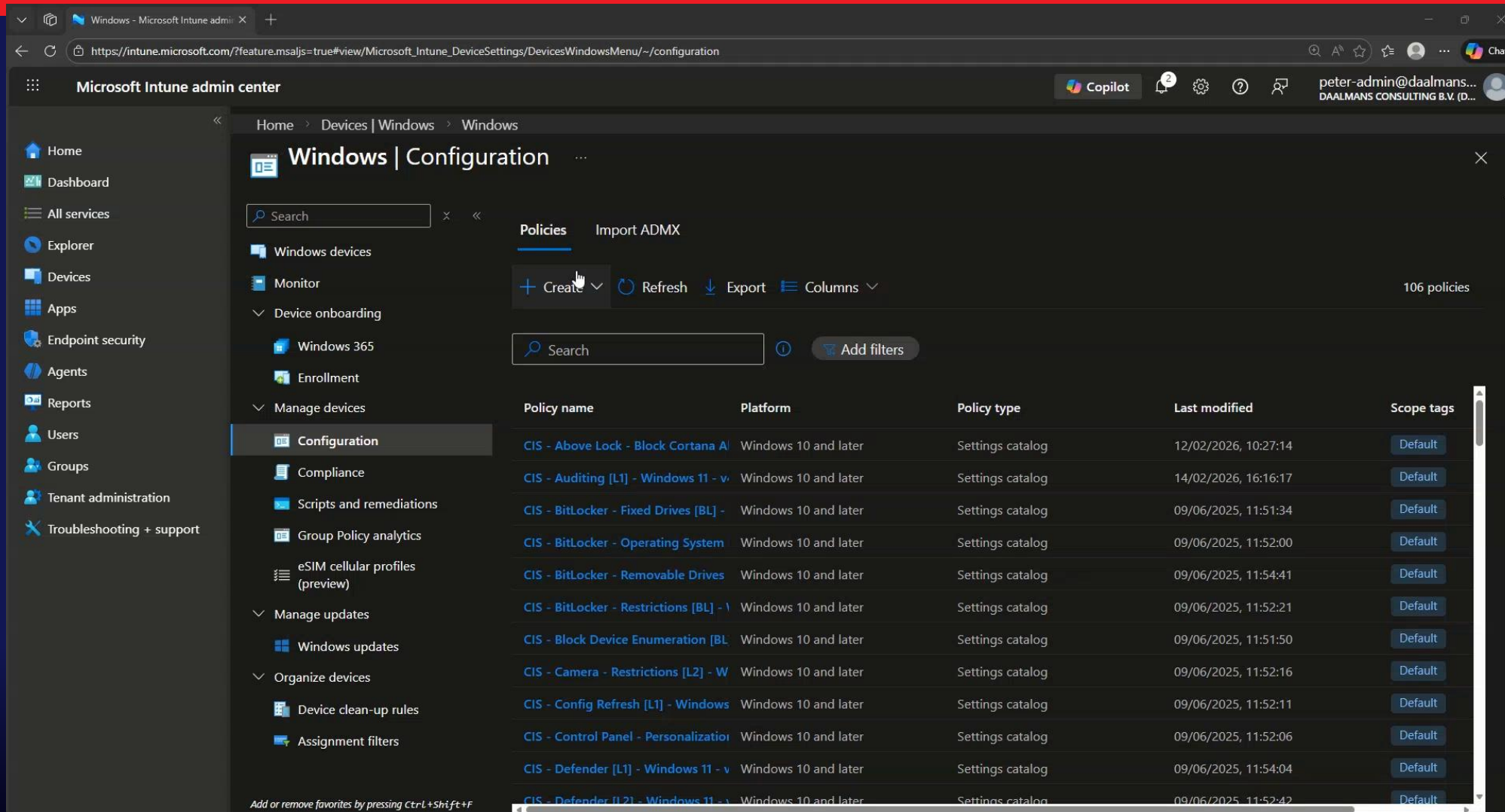
Boot-time Exploits

Vulnerabilities like BlackLotus targeting early boot

The expiration problem — why this matters now



Deploy via Microsoft Intune



The screenshot displays the Microsoft Intune admin center interface. The left sidebar contains navigation options: Home, Dashboard, All services, Explorer, Devices, Apps, Endpoint security, Agents, Reports, Users, Groups, Tenant administration, and Troubleshooting + support. The main content area is titled 'Windows | Configuration' and includes a search bar, a 'Policies' tab, and an 'Import ADMX' button. Below these are 'Create', 'Refresh', 'Export', and 'Columns' buttons. A table lists 106 policies, with columns for Policy name, Platform, Policy type, Last modified, and Scope tags. The table is filtered to show policies for Windows 10 and later.

Policy name	Platform	Policy type	Last modified	Scope tags
CIS - Above Lock - Block Cortana AI	Windows 10 and later	Settings catalog	12/02/2026, 10:27:14	Default
CIS - Auditing [L1] - Windows 11 - v	Windows 10 and later	Settings catalog	14/02/2026, 16:16:17	Default
CIS - BitLocker - Fixed Drives [BL] -	Windows 10 and later	Settings catalog	09/06/2025, 11:51:34	Default
CIS - BitLocker - Operating System	Windows 10 and later	Settings catalog	09/06/2025, 11:52:00	Default
CIS - BitLocker - Removable Drives	Windows 10 and later	Settings catalog	09/06/2025, 11:54:41	Default
CIS - BitLocker - Restrictions [BL] - 1	Windows 10 and later	Settings catalog	09/06/2025, 11:52:21	Default
CIS - Block Device Enumeration [BL]	Windows 10 and later	Settings catalog	09/06/2025, 11:51:50	Default
CIS - Camera - Restrictions [L2] - W	Windows 10 and later	Settings catalog	09/06/2025, 11:52:16	Default
CIS - Config Refresh [L1] - Windows	Windows 10 and later	Settings catalog	09/06/2025, 11:52:11	Default
CIS - Control Panel - Personalization	Windows 10 and later	Settings catalog	09/06/2025, 11:52:06	Default
CIS - Defender [L1] - Windows 11 - v	Windows 10 and later	Settings catalog	09/06/2025, 11:54:04	Default
CIS - Defender [L2] - Windows 11 - v	Windows 10 and later	Settings catalog	09/06/2025, 11:52:42	Default

Monitor Secure Boot certificates - Intune

Secure Boot status

Refresh Report Columns

7/10/2025

Search



Add filters

Up to date
3

Not up to date
2

Not applicable
2

Device count
7

Device name ↓	OS version ↑↓	Microsoft Entra device ID ↑↓	Secure Boot enabled ↑↓	Certificate status	Device model ↑↓
DESKTOP-6TTTGVA	10.0.26100.7840	02e250d6-21d4-4e72-b980-7296fce3cc5	Yes	✓ Up to date	Virtual Machine
DESKTOP-40DSE0H		7047604f-1ff5-4aad-9ecb-c47e9760c652	Unknown	? Not applicable	
DESKTOP-04L7EC9	10.0.26100.7840	023d96ae-7383-42cf-81c6-48d163cf858	Yes	✓ Up to date	Virtual Machine
DCL-VM01		fa474d0c-a0d6-4beb-9fe9-5f984e68390a	Unknown	? Not applicable	
DC-046666714157	10.0.26100.7922	f03e5db0-681c-4949-a876-4b9569a2607b	Yes	✗ Not up to date	Surface Laptop 4
DC-035296301257	10.0.26100.7781	644393e7-4eb2-49c3-8bea-4b635f7b00be	Yes	✗ Not up to date	Surface Book 3
CPC-pe-30F0P	10.0.26220.8062	eee65493-90bf-4afa-9ef4-e9702858462f	Yes	✓ Up to date	Virtual Machine

Why you might want to look into a different approach

- ◆ Known issue with the CSP path causing it to exit with error 65000
 - ◆ Policy is never effectively applied to the device

Enable Secureboot Certificate Updates

✖ Error

65000

- ◆ The device appears configured in Intune
 - ◆ but the underlying firmware-level change never takes place
- ◆ Output to assess and remediate is missing
- ◆ Secure boot reporting : latency / accuracy

Community is your friend

- ◆ Secure Boot Certificate Update (MindCore)
- ◆ Remediation script based solution
- ◆ Phased approach with detailed logging capabilities



Mattias Melkersen

STAGE	STATUS	WHAT IT MEANS
Stage 0	Secure Boot disabled	Updates cannot apply. Manual intervention required to enable Secure Boot in BIOS/UEFI, or convert legacy MBR disks to GPT.
Stage 1	Managed Opt-In not set	Not configured for managed certificate updates. The remediation script will automatically set the required registry key.
Stage 2	Configured, awaiting WU scan	Waiting for Windows Update. If stuck beyond the fallback limit (e.g. 30 days), the script will automatically use a direct method (WinCS) on its next run.
Stage 3	Update in progress	Windows Update is applying certificates. If failing with 0x80070002 (missing payload files), a new Cumulative Update must be installed.
Stage 4	CA2023 cert in UEFI DB	Certificate written to firmware, but device is still using the old boot manager. One reboot away from compliance.
Stage 5	✓ Compliant	The certificate transition is complete. The device is successfully booting from the 2023-signed boot manager.

Quick check

```
Administrator: C:\Program Files\PowerShell\7\pwsh.exe
PS C:\Windows\System32> Get-SecureBootSVN

FirmwareSVN      : 0.0
BootManagerSVN   : 8.0
StagedSVN        : 8.0
ComplianceStatus : Not compliant (Staged SVN does not match firmware SVN)
BootManagerPath  : \\.\HarddiskVolume1\EFI\Microsoft\Boot\bootmgfw.efi

PS C:\Windows\System32> Get-SecureBootUefi -Name KEK -decoded

SignatureOwner : 7facc7b6-127f-4e9c-9c5d-080f98994345
Subject        : CN=Lenovo Ltd. KEK CA 2012, O=Lenovo Ltd., L=Yokohama, S=Kanagawa, C=JP
Version        : 3
Algorithm      : sha256RSA
SerialNumber   : 00955243828A5A652E
ValidFrom      : 2012-06-29 05:35:34Z
ValidTo        : 2032-06-24 05:35:34Z

SignatureOwner : 77fa9abd-0359-4d32-bd60-28f4e78f784b
Subject        : CN=Microsoft Corporation KEK CA 2011, O=Microsoft Corporation, L=Redmond, S=Washington, C=US
Version        : 3
Algorithm      : sha256RSA
SerialNumber   : 610AD18800000000000003
ValidFrom      : 2011-06-24 15:41:29Z
ValidTo        : 2026-06-24 15:51:29Z

PS C:\Windows\System32>
```

* Requires Feb
Quality Update

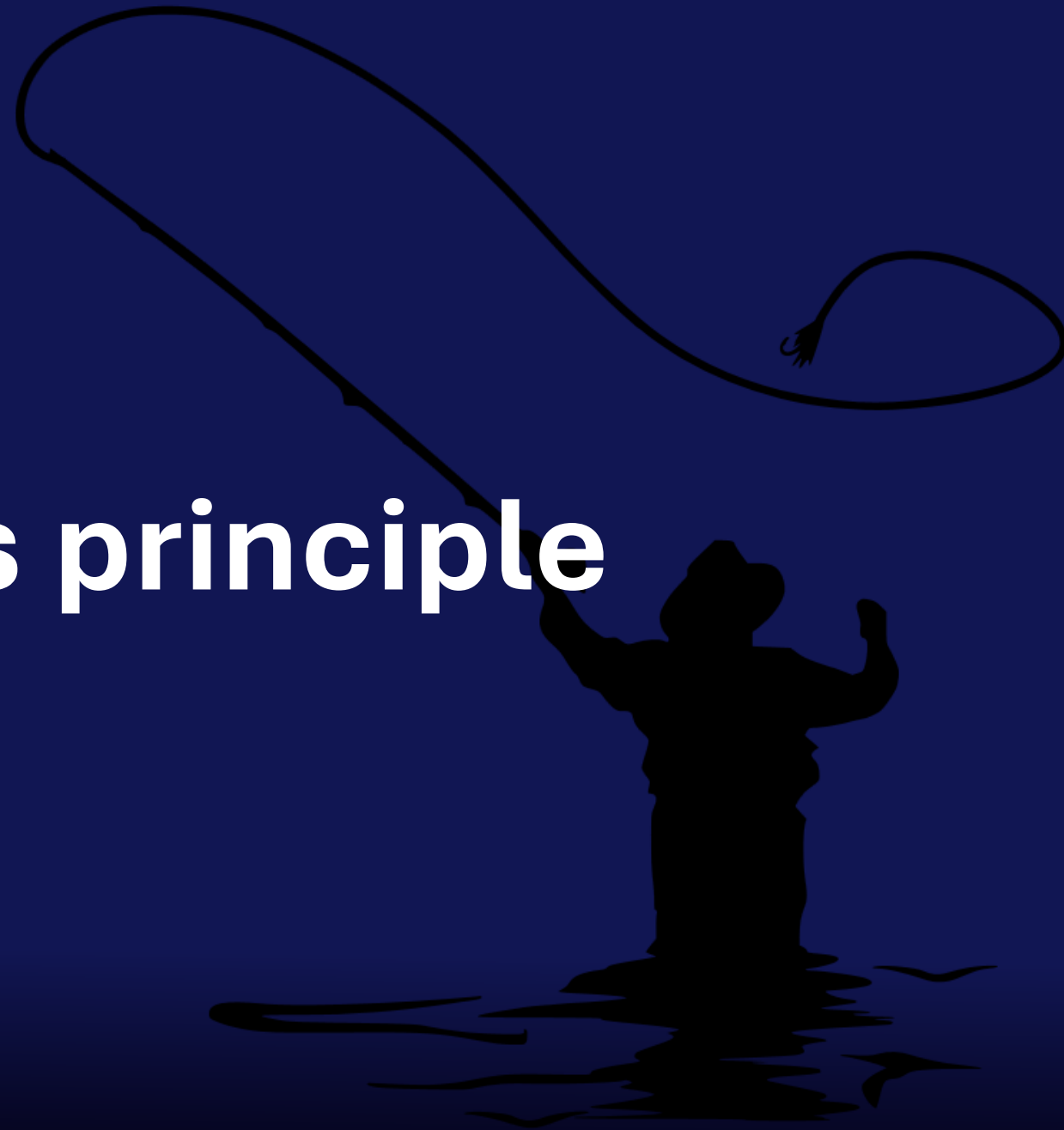
Tools: <https://github.com/gwblok/garytown/tree/master/BlackLotusKB5025885>

DEMO

Secure Boot



**Adopt four eyes principle
now..**

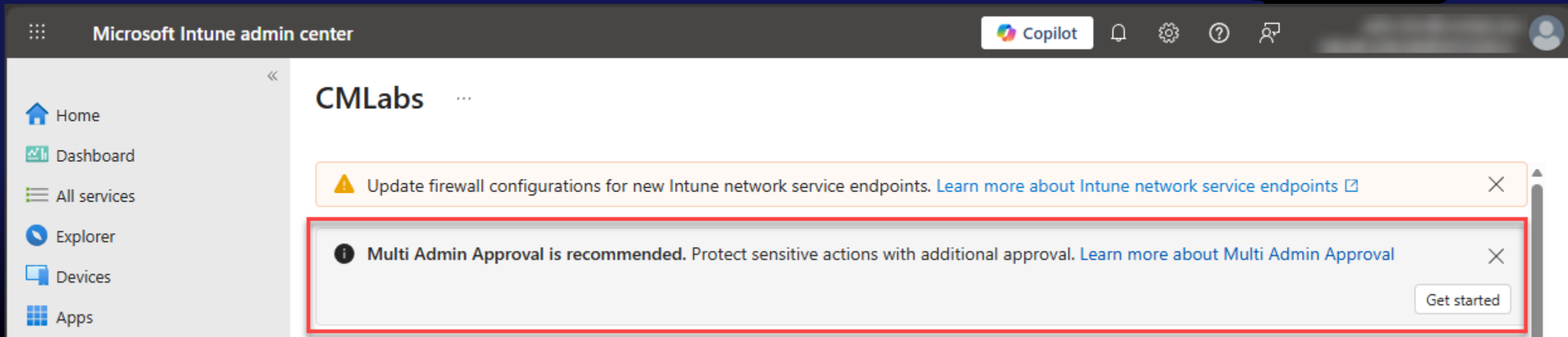




PART ONE: MULTI ADMIN APPROVAL

What is Multi Admin Account Approval

- ♦ *Intune access policies to require that a second administrative account is used to approve a change before the change is effectively applied*
- ♦ *Helps to protect against compromised admin accounts*
- ♦ *Only for select object types*



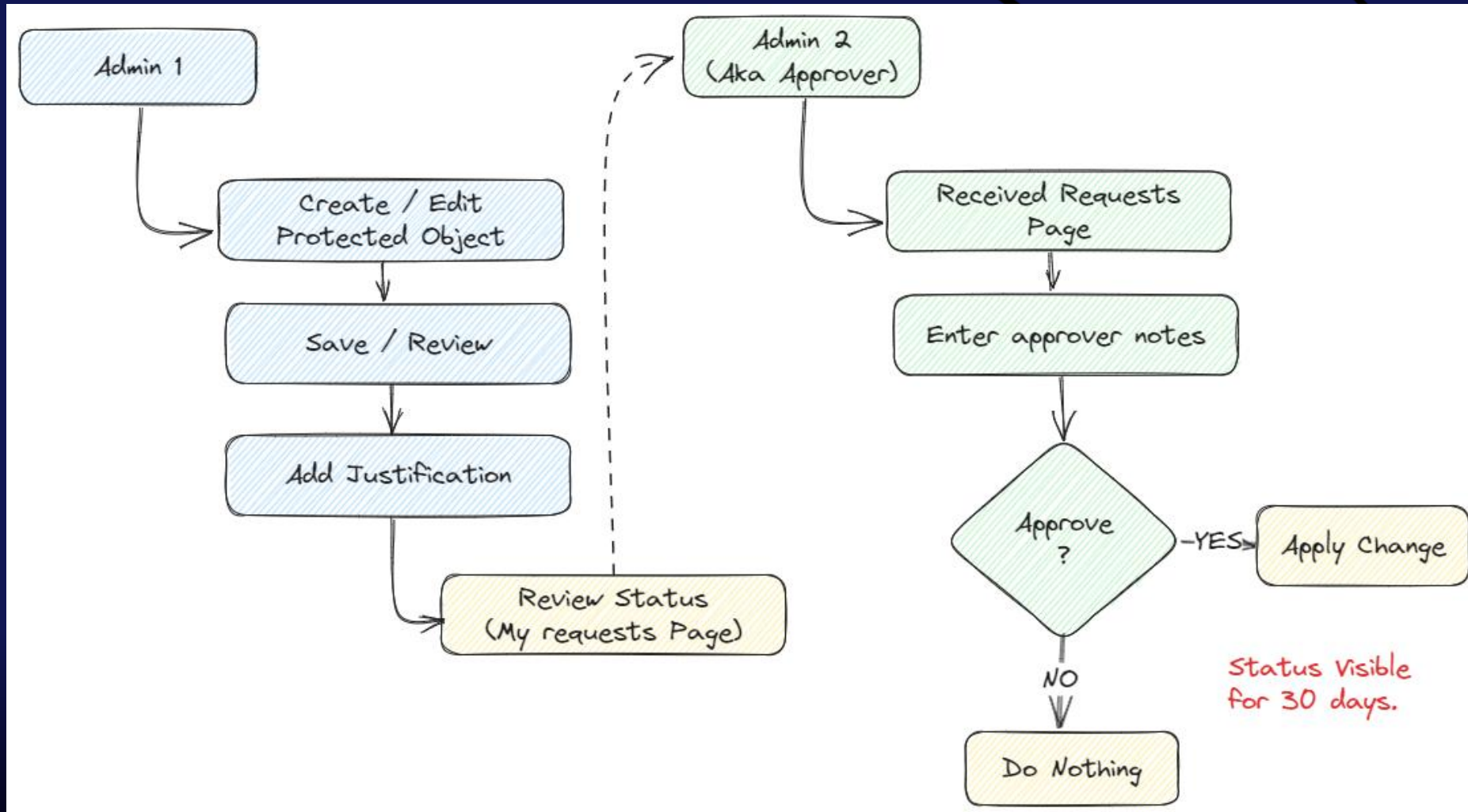
Prerequisites

- ♦ At least 2 administrator accounts
- ♦ No need for the Intune Service Administrator or Azure Global Administrator roles anymore



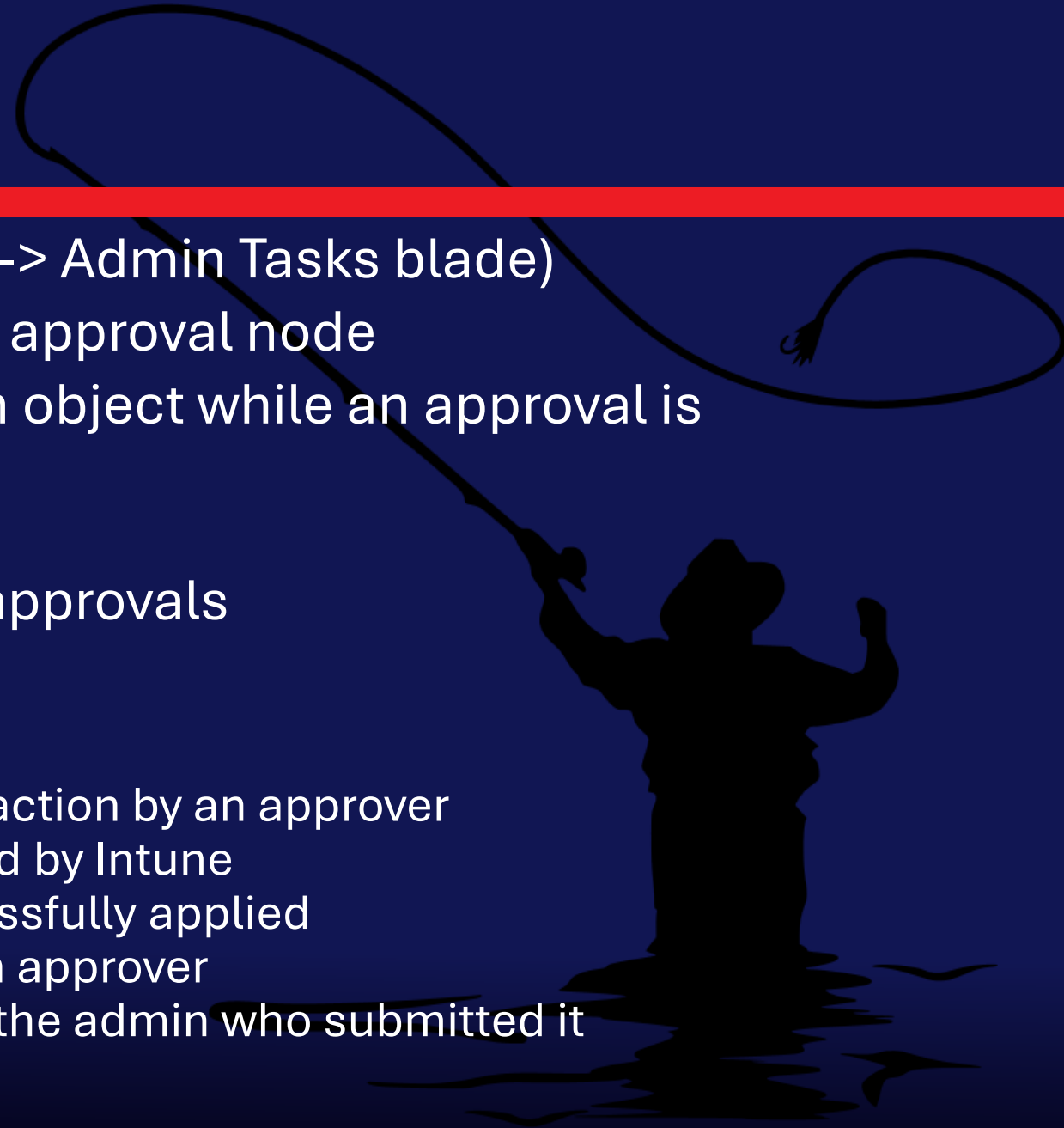
Multi Admin Approval	
Delete access policy ⓘ	☐ No ☐ Yes
Update access policy ⓘ	☐ No ☐ Yes
Read access policy ⓘ	☐ No ☐ Yes
Approval for Multi Admin Approval ⓘ	☒ No ☐ Yes
Create access policy ⓘ	☐ No ☐ Yes

Flow



Important

- ♦ Intune does not send notifications (-> Admin Tasks blade)
- ♦ Process required for monitoring the approval node
- ♦ No new requests are possible for an object while an approval is pending
- ♦ Intune Audit logs – for request and approvals
- ♦ Status conditions for a request:
 - ♦ Needs approval – Request is pending action by an approver
 - ♦ Approved – Request is being processed by Intune
 - ♦ Completed – Request has been successfully applied
 - ♦ Rejected – Request was rejected by an approver
 - ♦ Canceled – Request was canceled by the admin who submitted it



Multi admin workloads

Apps

Scripts

Compliance
policy

Configuration
policy

Device Delete

Device Retire

Device Wipe

(Intune) Role

Stryker incident review

Confirmed: On 11 Mar 2026, Stryker disclosed a cyber incident that caused global disruption to its Microsoft environment and affected access to systems supporting operations and corporate functions.

Confirmed: On 16 Mar, Stryker stated that the threat actor used Microsoft Intune to wipe its internal virtual infrastructure, rendering laptops, mobile devices, and virtual servers inoperable; Stryker also stated that this was not ransomware and that no malware was deployed.

Co

MAA & Review Agent better together



Multi Admin Approval (MAA)

A second Intune admin must approve high-risk actions before they execute

Initiator

Pending
Approval

Approved
& Applied

- Covered actions: deploy scripts, wipe devices, retire/delete policies
- Approval request visible in Intune portal
- Approver must have the same or higher RBAC permissions



Intune Review Agent (Security Copilot)

AI-powered analysis of Intune configurations — surfaces risks before they become incidents

Script Analysis

Deobfuscates and reviews PowerShell/shell scripts for malicious commands, C2 patterns, data exfil

Config Risk Review

Flags overly permissive policies, deprecated settings, and compliance gaps across profiles

Threat Explanation

Natural-language summaries of findings — tells you what's risky, why, and recommended remediation

Requires Microsoft Security Copilot (pay-as-you-go SCU capacity)

DEMO

Multi Admin Approval

With support of AI





IT ADMIN

SECURITY ADMIN

APP ADMIN

SUPPORT

PART TWO: RBAC IN MICROSOFT INTUNE

**NOT ALL WORKPLACE
ADMINS NEED TO BE
INTUNE ADMINISTRATOR!**



**ADOPT
LEAST PRIVILEGED ACCESS
NOW!**



RBAC and Scope Tags / permissions

- ♦ *Every role assignment combines three layers to control access in Intune*



Built-in roles cover most scenarios — custom roles enable least-privilege for your org

Scope – Making It Work

Scope (Groups) — WHO

Users and devices the admin manages: All Users, All Devices, or specific Entra ID groups

+

Scope Tags — WHAT

Objects the admin can see: policies, apps, config profiles

The Tag Rule: tag must exist on BOTH the role assignment AND the object — or the object is invisible

✓ Tags match → Visible

✗ Tag missing → Invisible

New objects get the "Default" tag — only visible to Default-scope admins

Privileged Access Management



No standing access — role expires automatically. Full audit trail in Entra logs.

Works with Entra built-in roles (e.g. Intune Administrator) and Privileged Access Groups

Privileged Access Groups in Intune RBAC

1

Create role-assignable group

Entra ID → New group → Azure AD roles can be assigned = Yes

2

Enable Privileged Access (PIM)

Group → Privileged Access → Enable → configure settings

3

Assign group to Intune RBAC role

Intune → Tenant admin → Roles → custom role → Assignments

4

Add eligible members

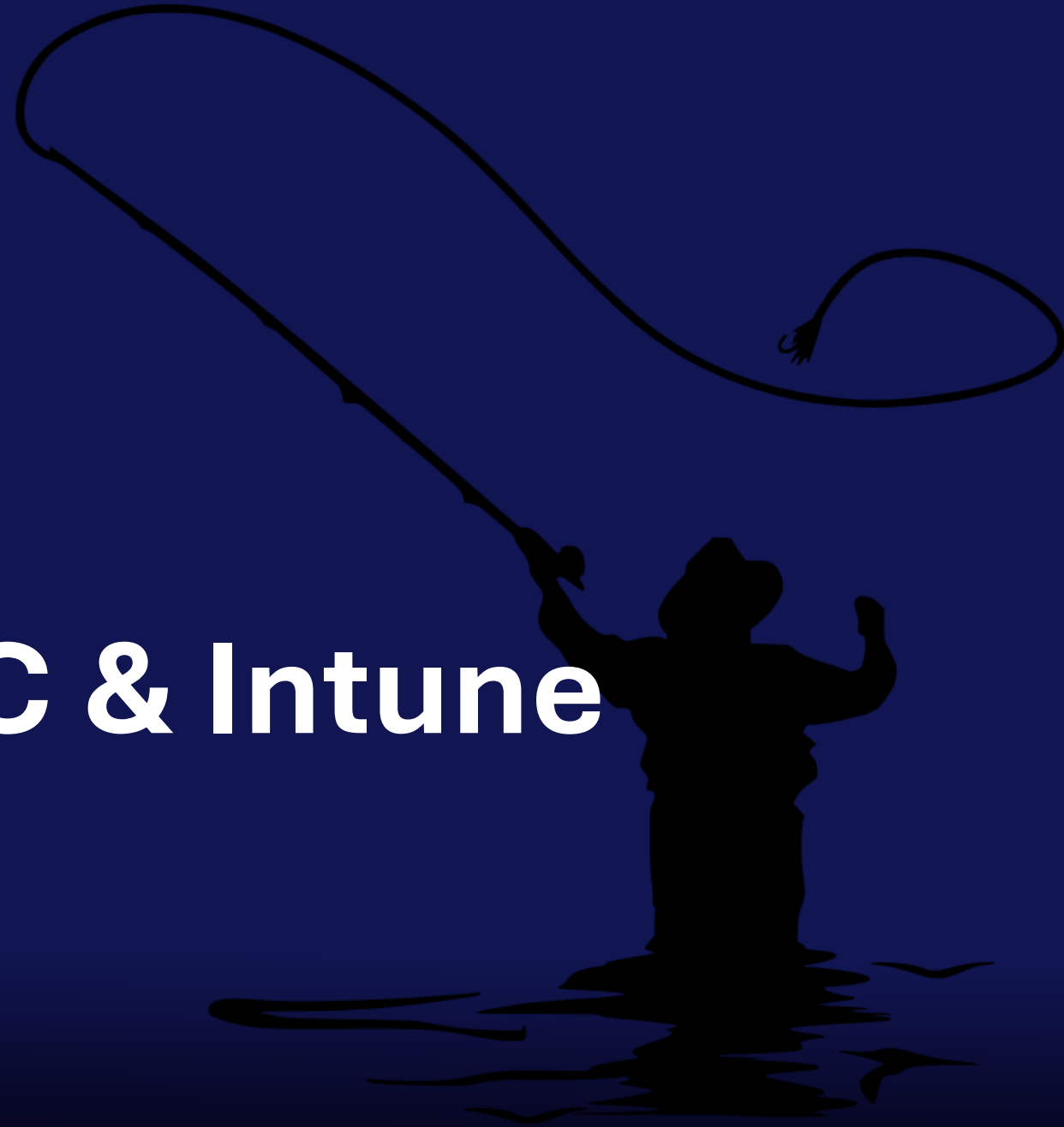
Entra PIM → group → Assignments → Eligible → select admins

⚠ Requires Entra ID P2 or Microsoft Entra ID Governance license



DEMO

JIT RBAC & Intune



EPM & Intune Agents



A silhouette of a cowboy wearing a hat, standing in water and throwing a lasso. The lasso is captured mid-air, forming a large loop. The background is a solid dark blue.

***Intune Suite and Security
Copilot will be part of E5***

Endpoint Privilege Management



Automatic and user-confirmed elevations

Insights based on elevation audits

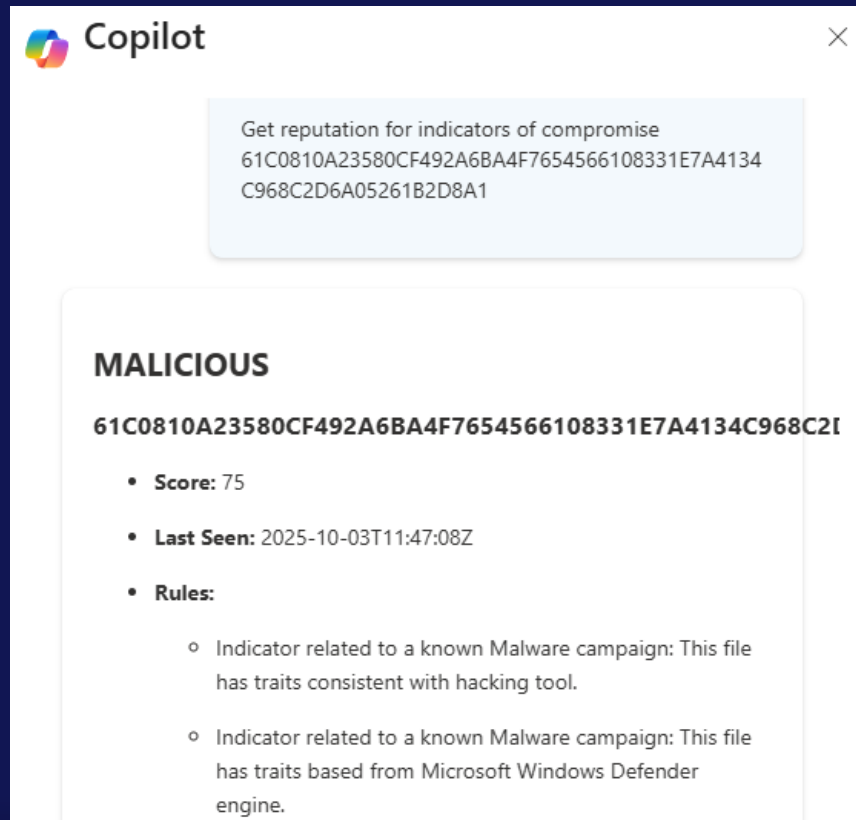
Rules based on organizational requirements

Easy addition or removal of rules

Tenant-level enablement, per device rollout

Now supports Windows 365

Use case – EPM & Copilot



- ◆ Copilot will check the file hash of an elevation against Microsoft Defender Threat Intelligence and provide any indicators of potential compromise

DEMO

EPM and Security Copilot better together

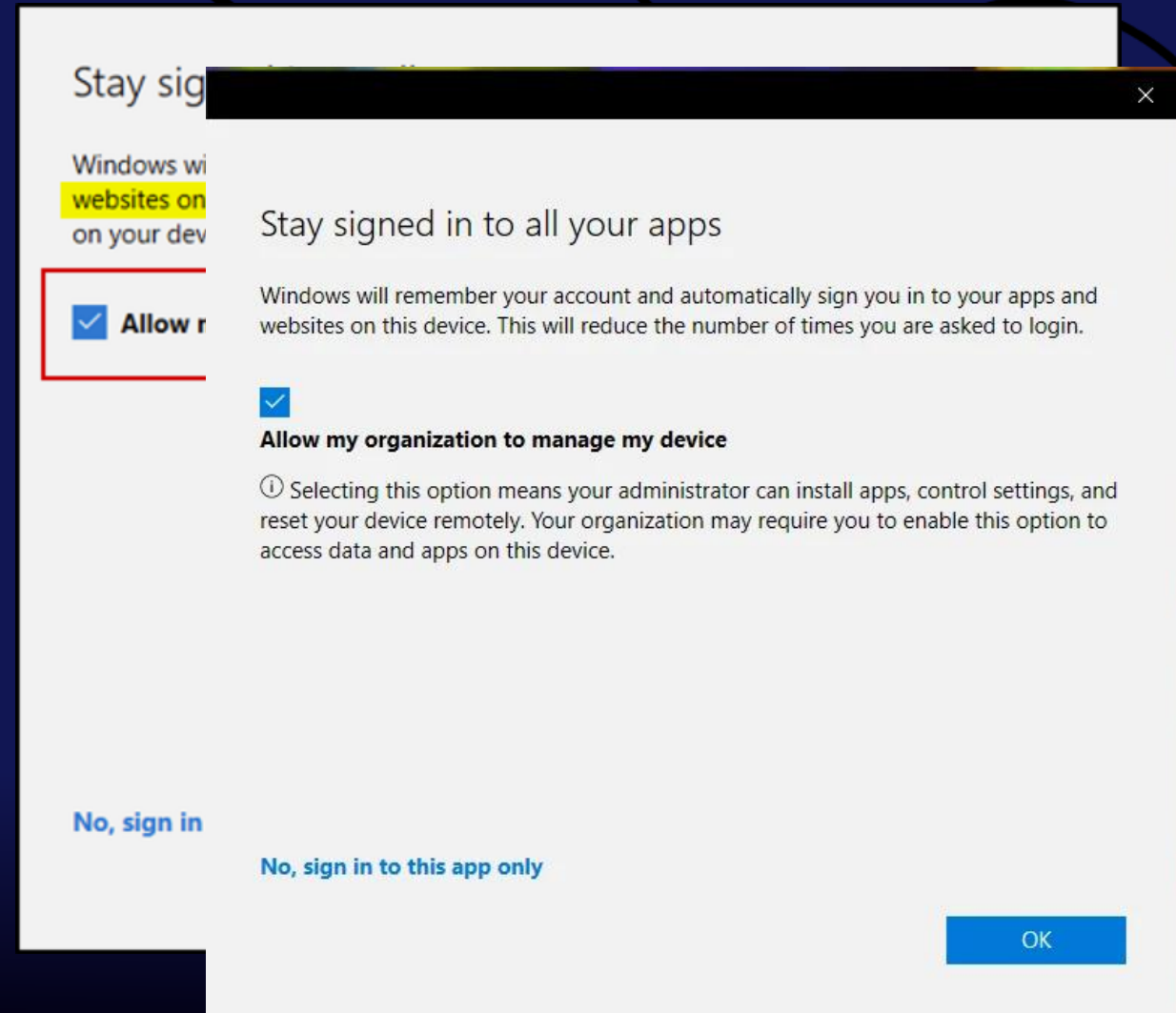


**Stop BYOD devices ending up
in your tenant**



The scenario ...

- ◆ When automatic enrollment is enabled in Intune, users who add a work or school account on Windows are automatically prompted to enroll their device into MDM
- ◆ Commonly occurs when signing in through any Microsoft 365 apps (Teams, Outlook, etc.) or other apps that require authentication
- ◆ Device registration and enrollment are combined in a single dialog box
- ◆ Users often miss the context of the prompt and leave the checkbox enabled (default)
- ◆ Device unknowingly gets enrolled and becomes managed by the IT department



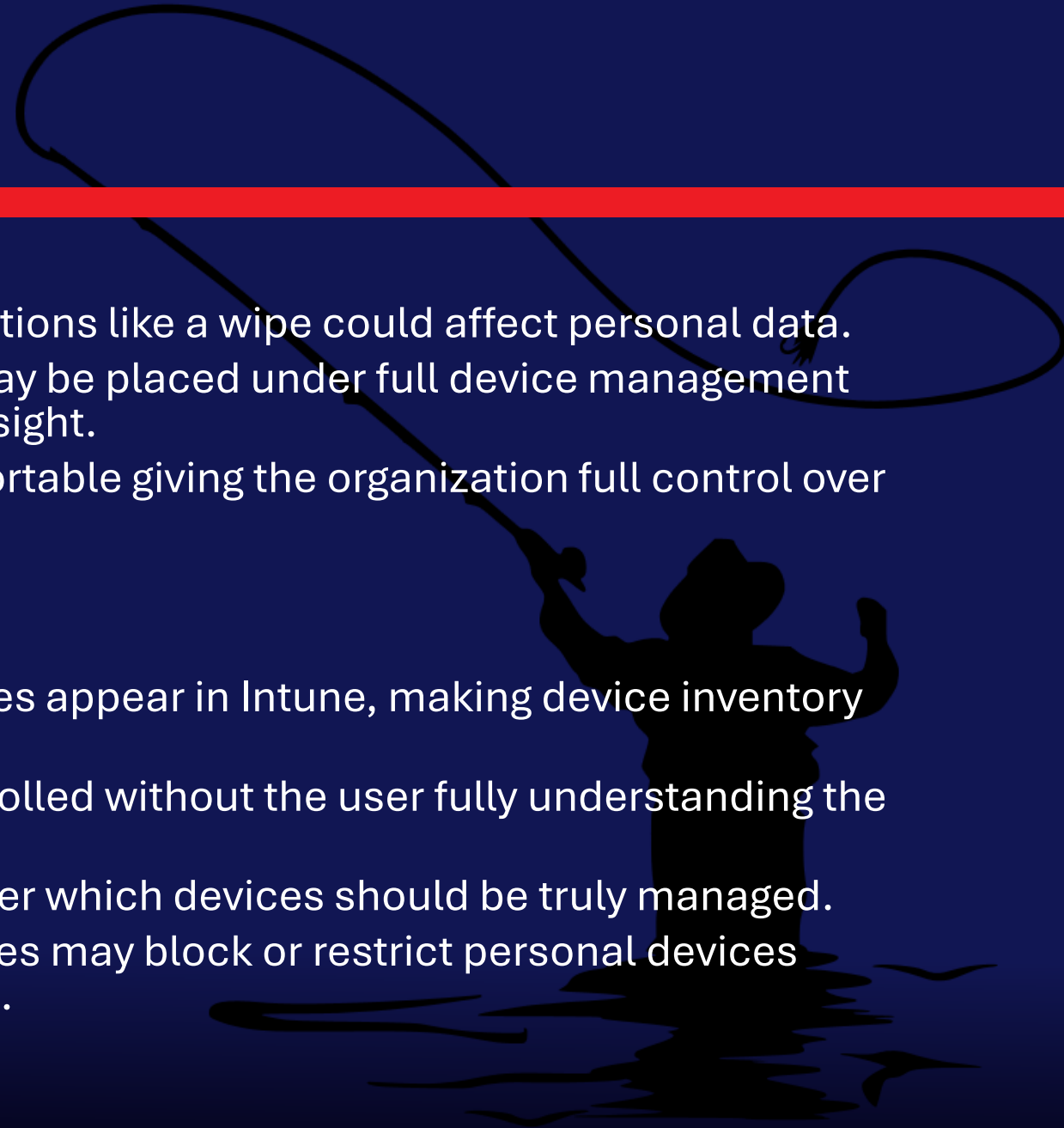
Why this matters ...

◆ Risks for Users

- ◆ Accidental full device wipe – Corporate actions like a wipe could affect personal data.
- ◆ Unwanted device management – Users may be placed under full device management even when only corporate apps need oversight.
- ◆ Privacy concerns – Users may be uncomfortable giving the organization full control over personal devices.

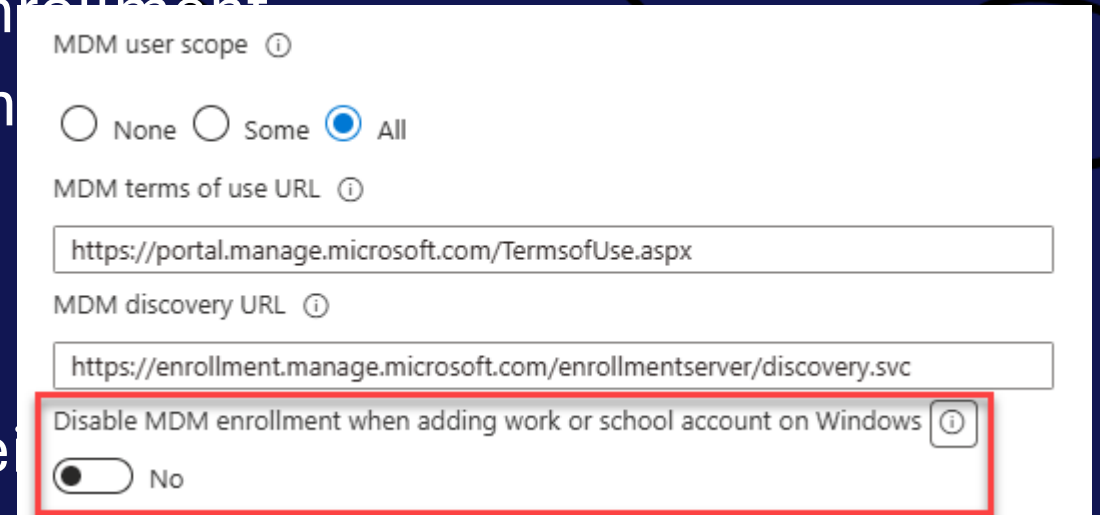
◆ Challenges for Admins

- ◆ Cluttered device records – Personal devices appear in Intune, making device inventory harder to manage.
- ◆ Unintended enrollments – Devices get enrolled without the user fully understanding the impact, creating compliance gaps.
- ◆ Limited flexibility – Admins lose control over which devices should be truly managed.
- ◆ Overreach of policies – Compliance policies may block or restrict personal devices unnecessarily, complicating management.



New setting brings more control ...

- ◆ Devices > Enrollment > Automatic Enrollment
- ◆ Disable MDM enrollment when adding Windows option
- ◆ Disabled by default
- ◆ Does not apply when a user adds the
 - ◆ Windows Settings flow
 - ◆ Microsoft Intune Company portal
 - ◆ Applications that explicitly require device enrollment as a prerequisite for access
- ◆ Only affects the account registration flow when adding a work or school account via apps like Microsoft 365 Apps or Edge.



MDM user scope ⓘ

☐ None ☐ Some ☒ All

MDM terms of use URL ⓘ

MDM discovery URL ⓘ

Disable MDM enrollment when adding work or school account on Windows ⓘ

☐ No

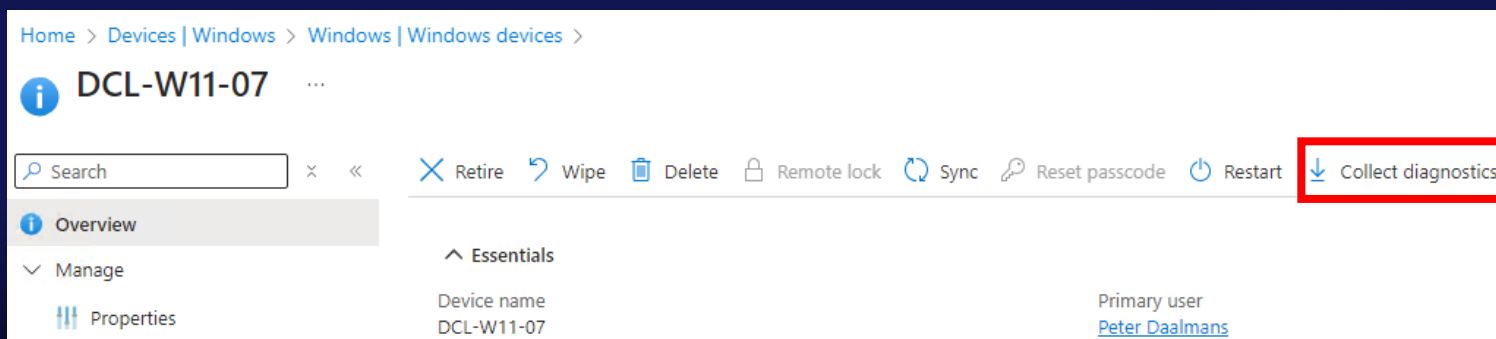
Gathering custom log files



How to retrieve custom logs files file

- ◆ Include custom logs in diagnostics zip

```
msiexec /I "filename.msi" /qn /I*v  
%windir%\ccm\Logs\logfile-application-  
version.log
```



- %ProgramData%\Microsoft\DiagnosticLogCSP\Collectors*.etl
- %ProgramFiles%\Microsoft EPM Agent\Logs*.*
- %ProgramData%\Microsoft\IntuneManagementExtension\Logs*.*
- %ProgramData%\Microsoft\Windows Defender\Support\MpSupportFiles.cab
- %ProgramData%\Microsoft\Windows\WlanReport\wlan-report-latest.html
- %ProgramData%\USOShared\logs\system*.etl
- %ProgramData%\Microsoft Update Health Tools\Logs*.etl
- %temp%\CloudDesktop*.log
- %temp%\MDMDiagnostics\battery-report.html
- %temp%\MDMDiagnostics\energy-report.html
- %temp%\MDMDiagnostics\mdmlogs- <Date/Time> .cab
- %temp%\MDMDiagnostics\msinfo32.log
- %windir%\ccm\logs*.log
- %windir%\ccmsetup\logs*.log
- %windir%\logs\CBS\cbs.log
- %windir%\logs\measuredboot*.*
- %windir%\logs\Panther\unattendgc\setupact.log
- %windir%\logs\SoftwareDistribution\ReportingEvent\measuredboot*.log
- %windir%\Logs\SetupDiag\SetupDiagResults.xml
- %windir%\logs\WindowsUpdate*.etl
- %windir%\SensorFramework*.etl
- %windir%\system32\config\systemprofile\AppData\Local\mdm*.log
- %windir%\temp%computername%.log
- %windir%\temp\officeclicktorun*.log
- %TEMP%\winget\defaultstate*.log

DEMO

Gathering custom logfiles

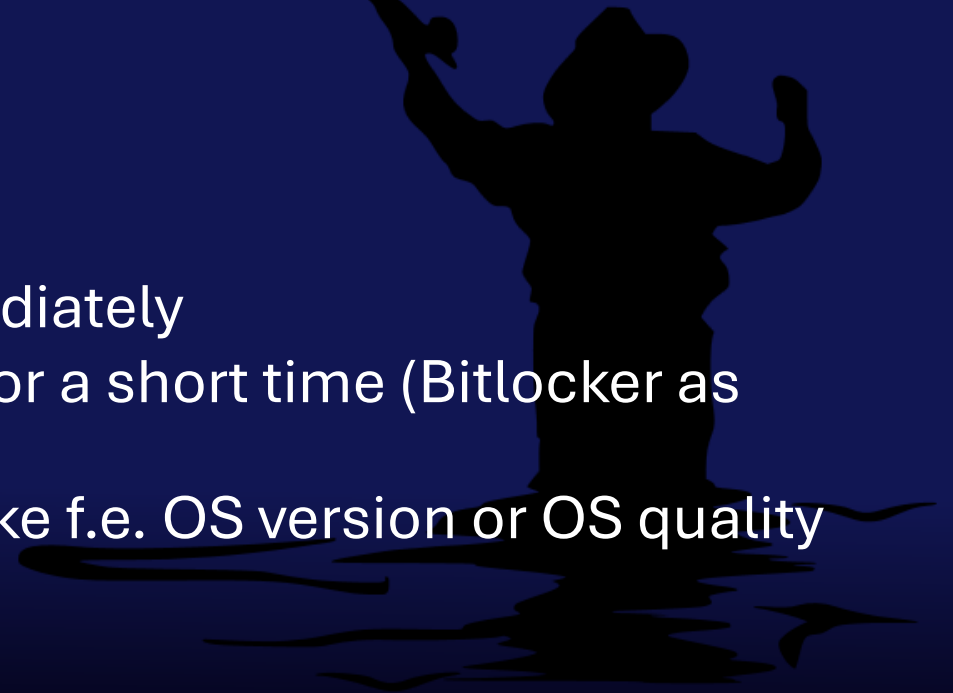


Compliance Policies



Compliance Policies – One versus Many

- ◆ One versus Many
- ◆ Built your Compliance Policies based on your grace period timeout requirements
- ◆ Did you know that you can also use 0.5 days as a way to say that the grace period is ½ day
- ◆ Create different compliance policy for:
 - ◆ Things you want to be non-compliant immediately
 - ◆ Things you want to be non-compliant only for a short time (Bitlocker as part of device health)
 - ◆ Things which you want to give some time, like f.e. OS version or OS quality update



DEMO

Compliance policies



Admin Tasks



Admin tasks ... in a nutshell

- ◆ One pane of glass for high-impact admin work
 - ◆ Found at Tenant administration > Admin tasks
 - ◆ Centralized queue of approvals and remediation work — no more clicking across nodes
- ◆ Brings together tasks from:
 - ◆ **Endpoint Privilege Management** — file elevation requests
 - ◆ **Defender for Endpoint** — security tasks & remediation
 - ◆ **Multi Admin Approval** — scripts, remote actions, role changes, wipes
 - ◆ **Device Offboarding Agent** (preview, requires Security Copilot) — stale device cleanup
- ◆ Things to know
 - ◆ RBAC scoped — you only see tasks your assigned roles permit (needs Organization > Read)
 - ◆ Selecting a task opens the same pane as the source node — workflows unchanged
 - ◆ Statuses: Active, Pending, Completed, Rejected, Expired, Needs approval (expired tasks auto-purge after 30 days)

Admin Tasks

Search ? Add filters						
Task	Source	Status	Assigned to	Due in	Last updated	Created
mimikatz.exe	Endpoint privilege management approval	⚠ Pending	Security groups	1 day	04/27/2026, 05:12 PM	04/27/2026, 05:12 PM
test	Multi admin approval	✖ Rejected	Security groups	Overdue	05/01/2026, 01:34 AM	04/27/2026, 05:12 PM
Just another script	Multi admin approval	✖ Cancelled	Security groups	Overdue	05/01/2026, 01:34 AM	04/27/2026, 05:12 PM
MAA - Scripts	Multi admin approval	✓ Completed	Security groups	Overdue	05/01/2026, 01:34 AM	04/27/2026, 05:12 PM
Update Adobe Acrobat Reader Dc	Defender security task	✓ Active	Security groups	Overdue	01/05/2023, 04:02 PM	01/05/2023, 04:02 PM
Update Citrix Workspace	Defender security task	✓ Active	Security groups	Overdue	04/30/2020, 11:14 AM	04/30/2020, 11:14 AM
Update Chrome	Defender security task	⚠ Pending	Security groups	Overdue	11/05/2019, 01:17 PM	11/05/2019, 01:17 PM

Browser extensions



Why you should care ...

NEWS 14 April 2026

Malicious Chrome Extensions Campaign Exposes User Data

Some ChatGPT browser extensions are stealing your data

A threat actor
session tokens

MALWARE

That AI Extension Helping You Write Emails? It's Reading Them First

MALWARE & THREATS

Over 300 Malicious Chrome Extensions Caught Leaking or Stealing User Data

on combined downloads, the extensions expose users to tracking and data theft.

Research • March 5 • 6 min read

Malicious AI Assistant Extensions Harvest LLM Chat Histories

By [Microsoft Defender Security Research Team](#)

Millions of people spied on by malicious Chrome and Edge



How to get control over extensions used ...

- ◆ A single configuration profile is (or can be) sufficient

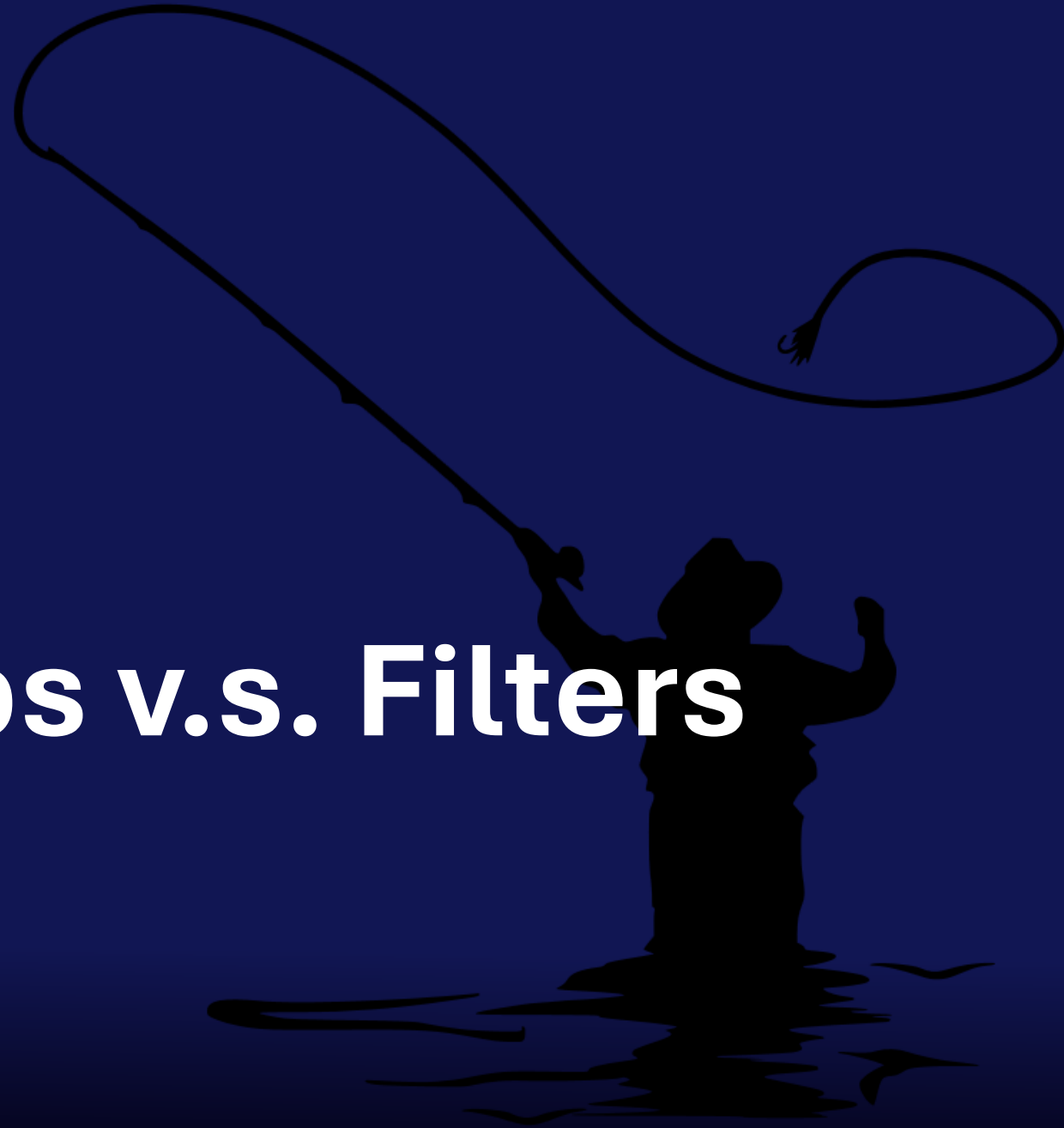
Extensions		
Extension IDs the user should be prevented from installing (or * for all) (Device)	*	
Allow specific extensions to be installed ①	Enabled	
Extension IDs to exempt from the block list (Device)	mjifldffjokfhokbjanjgmeabmhflb	
Control which extensions are installed silently ①	Enabled	
Extension/App IDs and update URLs to be silently installed (Device)	jbkfoedolllekghcbcoahefnbanhhlh, odfafepnkmhbccpbejgmiehpchacaeak, ifoakfbpdcdoeenechcleahebpibofpc	
Control which extensions cannot be installed ①	Enabled	

Block users from installing any extensions

Whitelist the ones you allow to be installed

Force install extensions you want to be installed

Dynamic Groups v.s. Filters



Dynamic Groups v.s. Filters

- ♦ Dynamic Groups can be useful
 - ♦ No guarantee **when** they calculate
- ♦ Filters
 - ♦ Calculate on the device itself
 - ♦ When used with All Users/All Devices you are limited to other assignments
- ♦ Be careful with filters though, you cannot query on properties you don't receive. So always check

Example:

(device.enrollmentProfileName -startsWith "AE - Corporate-owned, fully managed device profile - v")

versus

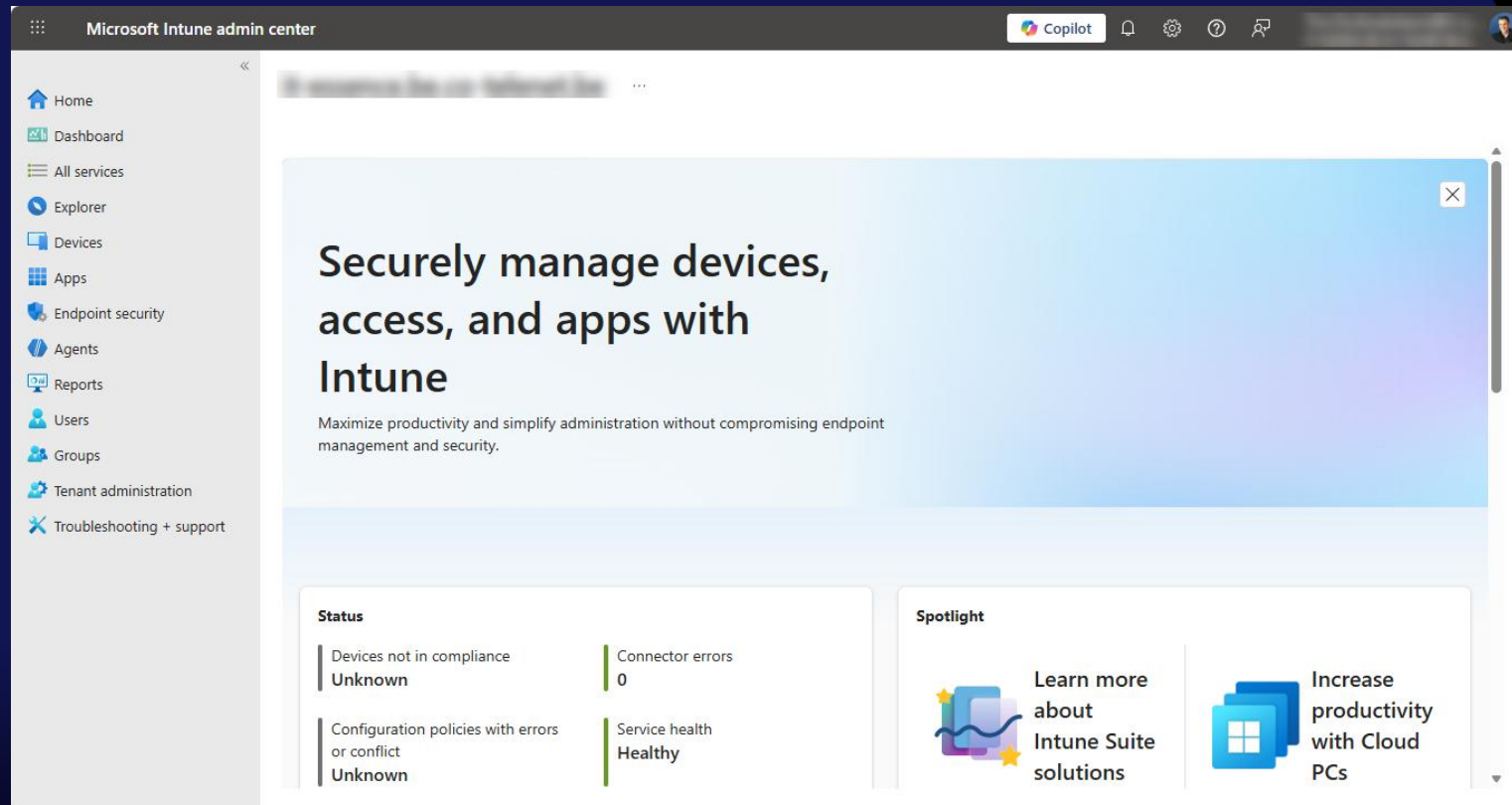
(device.enrollmentProfileName -startsWith "AE - Corporate-owned, fully managed device profile - v") and (device.enrollmentProfileName -ne \$null)

Portals Overload



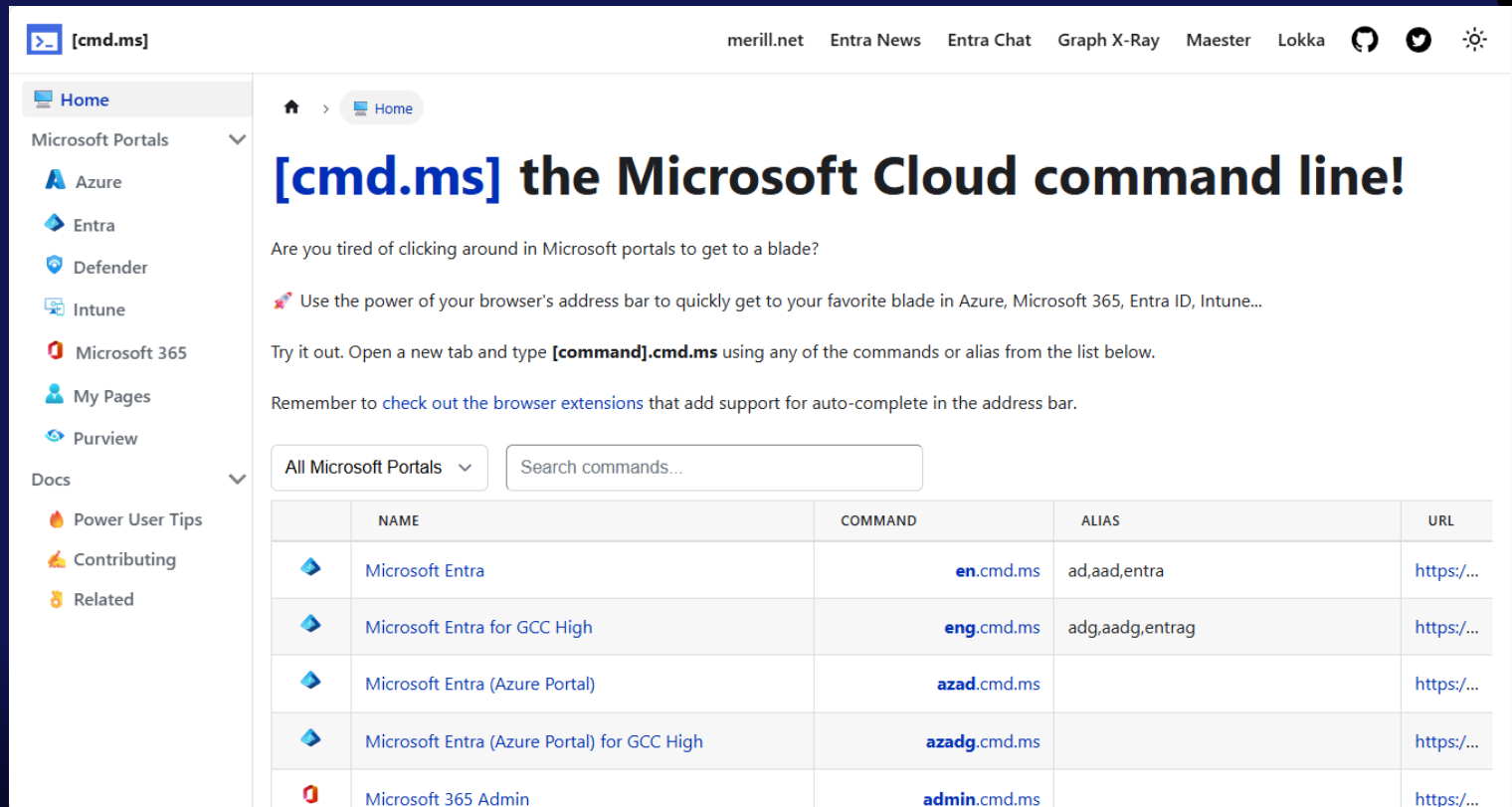
One is simply not enough ...

Intune.Microsoft.com



The quicker way to navigate : cmd.ms

Command Line alike access to consoles



[cmd.ms] the Microsoft Cloud command line!

Are you tired of clicking around in Microsoft portals to get to a blade?

Use the power of your browser's address bar to quickly get to your favorite blade in Azure, Microsoft 365, Entra ID, Intune...

Try it out. Open a new tab and type **[command].cmd.ms** using any of the commands or alias from the list below.

Remember to [check out the browser extensions](#) that add support for auto-complete in the address bar.

All Microsoft Portals

	NAME	COMMAND	ALIAS	URL
	Microsoft Entra	en.cmd.ms	ad,aad,entra	https://...
	Microsoft Entra for GCC High	eng.cmd.ms	adg,aadg,entrag	https://...
	Microsoft Entra (Azure Portal)	azad.cmd.ms		https://...
	Microsoft Entra (Azure Portal) for GCC High	azadg.cmd.ms		https://...
	Microsoft 365 Admin	admin.cmd.ms		https://...

Recently revamped

[Docs](#) [Power User Tips](#) [merill.net](#) [Entra News](#) [Entra Chat](#) [Graph X-Ray](#) [Maester](#) [Lokka](#) [GitHub](#)

CMD.MS

The Microsoft Cloud command line

Type a command to jump to any Microsoft portal – or use `[command].cmd.ms` in your browser address bar.

🚀 Use the power of your browser's address bar, check out the [Power User Tips](#)

C:\cmd.ms> Search commands...

355 commands found

COMMAND	NAME	ALIAS
enpim.cmd.ms	PIM	adpim , pim
en.cmd.ms	Microsoft Entra	ad , aad , entra
enusers.cmd.ms	Users	adusers
engroups.cmd.ms	Groups	adgroups
enca.cmd.ms	Conditional Access	adca
enapps.cmd.ms	Enterprise applications	adentapp , adapp , adapps
enappreg.cmd.ms	App registrations	adappreg
in.cmd.ms	Intune	intune , endpoint

⬆

 Navigate

Enter

 Open

/

 Search

Esc

 Clear

Theme

?

 Help

Made in 🇦🇺 Australia with ❤️

DEMO

Portals Overload

Quick demo on cmd.ms



Endpoint Analytics



A silhouette of a person standing in water, casting a lasso. The lasso is in mid-air, forming a large loop. The person is holding the handle of the lasso with both hands.

***Why do you want to look at
Endpoint Analytics?***

Because you want

.... to be proactive

.... to have insights about

- how your apps are doing
- How your devices are performing

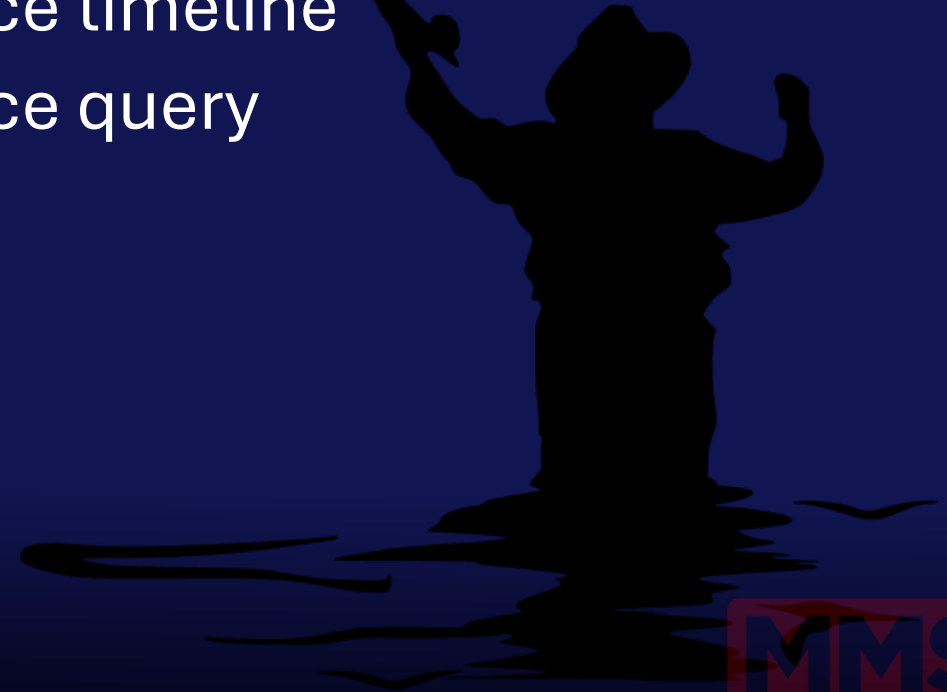
Its not only Intune Suite, but a lot is also already there

Already in Intune

- ◆ Startup performance
 - ◆ Model performance
 - ◆ Device performance (incl BSOD)
 - ◆ Startup process info
 - ◆ Restart info
- ◆ Application reliability
 - ◆ App Crashes
 - ◆ App Performance
 - ◆ Model performance
- ◆ Model/device score

Added by Intune Suite

- ◆ Battery health
- ◆ Resource performance
- ◆ Device timeline
- ◆ Device query



App name ↑↓	App publisher ↑↓	Active devices (14 days)	① ↑↓	App reliability score	① ↑↓	Total usage durat... ↑↓	Total crashes (14 ... ↑↓	Mean time to failure ①
OUTLOOK.EXE	Microsoft Corporation	864		67		10554 hours, 59 minut...	303	34 hours, 50 minutes
		210		16		44 hours, 14 minutes	103	25 minutes
LogonUI.exe		510		0		85 hours, 27 minutes	85	1 hour
		358		5		243 hours, 12 minutes	79	3 hours, 4 minutes
AcroRd32.exe	Adobe Systems Incorp...	618		28		731 hours, 7 minutes	67	10 hours, 54 minutes
WINWORD.EXE	Microsoft Corporation	782		100		5115 hours, 10 minutes	59	86 hours, 41 minutes
RtkAudUService64.exe	Realtek Semiconductor	333		50		0 minutes	20	0 minutes

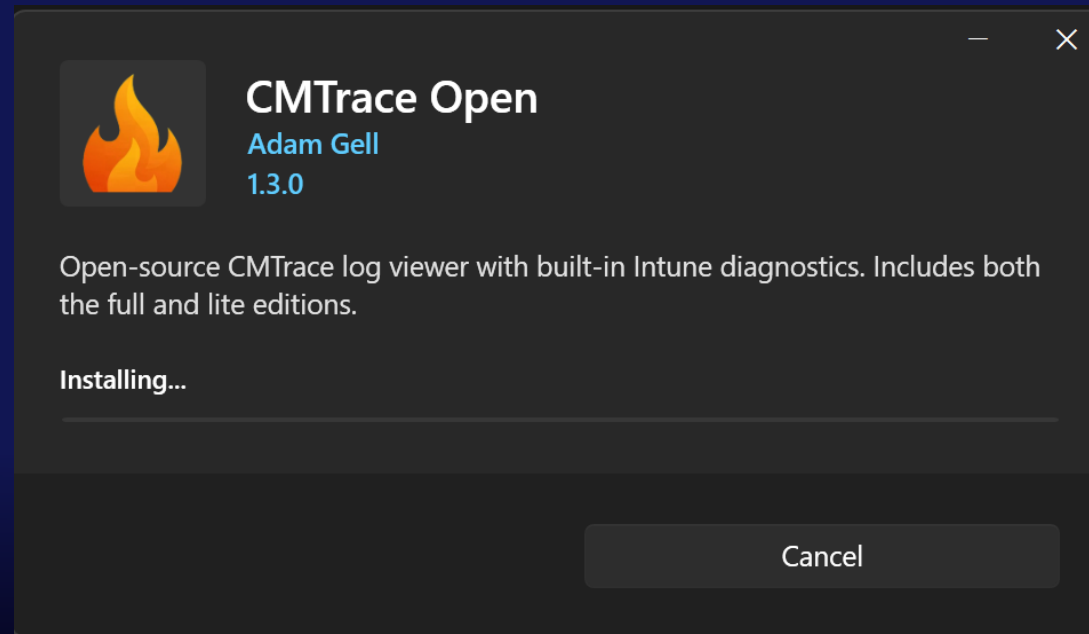
Troubleshooting

CMTrace is dead ... long live CMTrace



Community is your friend

- CMTrace Open
- Community tool inspired by the original CMTrace
- [GitHub - adamgell/cmtraceopen](https://github.com/adamgell/cmtraceopen)



Adam Gell

DEMO

CMTrace Open



A silhouette of a person standing in water, casting a long, curved net. The net is in mid-air, forming a large loop. The person is wearing a hat and a long-sleeved shirt. The background is a solid dark blue.

Keeping up in an ever evolving world

Helpful resources ...

♦ Social Media – X

- ♦ @MSIntune
- ♦ @IntuneSupportTeam (Intune support team)
- ♦ MVP's ... many are on the lists maintained by Jason Sandys
- ♦ And many more ...

♦ Intune Blog

<https://techcommunity.microsoft.com/t5/microsoft-intune-blog/bg-p/MicrosoftEndpointManagerBlog>

♦ What's New

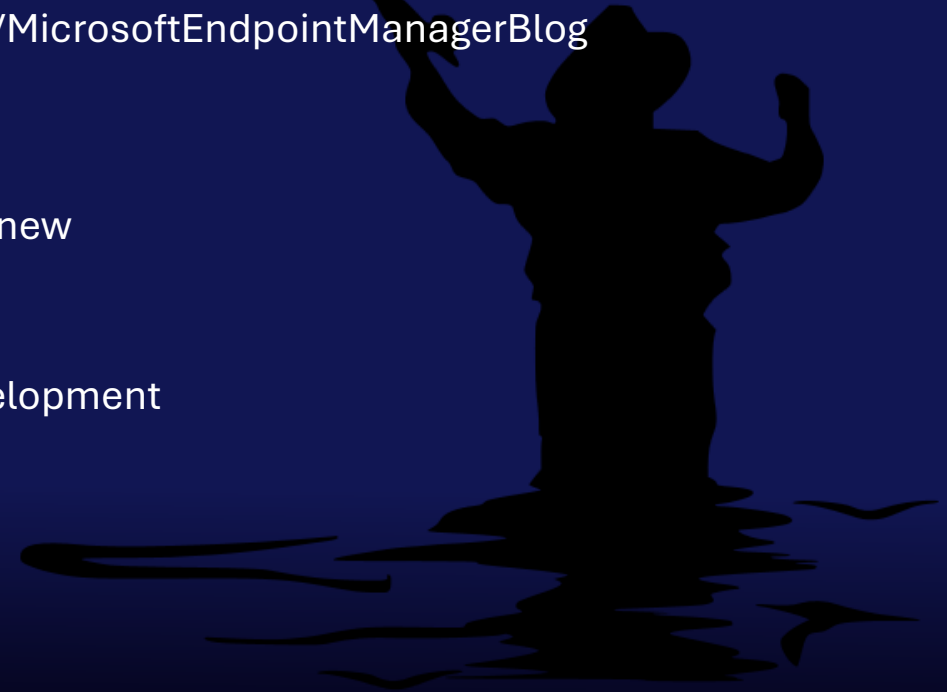
<https://docs.microsoft.com/en-us/mem/intune/fundamentals/whats-new>

♦ What is in development

<https://docs.microsoft.com/en-us/mem/intune/fundamentals/in-development>

♦ Putting it all together

<https://updates.getcurrent.cloud>



SAVE THE DATES

Oct 25-28, 2026



May 2-6, 2027



Oct 10-13, 2027



Extended Q&A



Recast



ninjaOne®



numecent®



Secure Boot

Certificate Expiration

What IT Administrators Need to Know & Do

- 15-year certificate lifecycle ending
- Affects all Secure Boot-enabled Windows devices
- Intune Remediation: the proven deployment path

Based on: Mindcore Techblog | Microsoft IT Pro Guidance



EXPIRES JUNE 2026

ACTION REQUIRED NOW

What Is Secure Boot?

The Foundation of Boot-Time Trust

Secure Boot is a UEFI firmware security feature that validates every piece of software in the boot chain before it runs.

It uses a Public Key Infrastructure (PKI) model where Microsoft and OEM Certificate Authorities (CAs) issue trusted digital certificates stored in the device firmware.

During startup, each component — firmware drivers, bootloaders, option ROMs — must present a valid signature. Any unsigned or tampered code is blocked before Windows even loads.

Threats Secure Boot Prevents

Bootkits & Rootkits

Malware targeting bootloader or kernel at startup

Firmware Attacks

Tampering or replacing UEFI firmware components

Code Injection

Unauthorized code injected into the boot sequence

Unsigned Kernel Modules

Loading tampered or unverified kernel modules

Boot-time Exploits

Vulnerabilities like BlackLotus targeting early boot

The Expiration Problem: Why This Matters Now

EXPIRING: June 2026

**Microsoft Corp.
KEK CA 2011**

Signs DB/DBX updates

EXPIRING: June 2026

**Microsoft UEFI
CA 2011**

3rd-party bootloaders & ROMs

EXPIRING: October 2026

**Windows Production
PCA 2011**

Signs Windows bootloader

REPLACEMENT (2023)

KEK CA 2023

REPLACEMENT (2023)

**UEFI CA 2023 &
Option ROM CA 2023**

REPLACEMENT (2023)

**Windows UEFI
CA 2023**

 Without the new 2023 certificates: devices lose the ability to receive Secure Boot security updates · cannot trust third-party software signed with new certs · boot manager fixes stop arriving · exposure to BlackLotus-style attacks grows over time.

Who Is Affected? Scope & Impact

IN SCOPE

- **Windows 11 Enterprise / Pro**
Intune managed — primary target
- **Windows 10 Enterprise / Pro**
Intune managed (ensure ESU is active)
- **Azure Virtual Machines**
VMs with Trusted Launch / vTPM enabled
- **Azure Virtual Desktop**
Session hosts with Secure Boot enabled
- **Co-managed devices**
Via Intune workload assignment

OUT OF SCOPE

- **BYOD / Personal devices**
Rely on standard Windows Update rollout
- **WSUS-only / air-gapped devices**
Need direct registry or GPO approach
- **Devices with Secure Boot OFF**
Requires manual firmware intervention
- **Windows Server**
Separate process — requires manual action
- **Legacy BIOS (MBR) devices**
No Secure Boot support at all

The Device Update Journey: 5 Stages

0

Stage 0: Secure Boot OFF

Legacy BIOS/MBR or Secure Boot disabled.
Manual firmware action required.

1

Stage 1: Opt-In Needed

Secure Boot ON. Registry key
MicrosoftUpdateManagedOptIn not yet set.
Remediation script writes the key.

2

Stage 2: Opted In

Key set (0x5944). Device awaits Windows
Update to stage the new 2023 certificate
payload.

3

Stage 3: KEK Updated

New KEK CA 2023 certificate installed in UEFI.
DB update is next.

4

Stage 4: Cert Staged

Windows UEFI CA 2023 in DB. Reboot required
to activate. Hotpatch devices may sit here.

5

Stage 5: Compliant ✓

All new 2023 certs active. Device fully protected
and future-proofed.

Deployment Approach: Intune Remediations

⚠ Why not Intune CSP profiles? The native Secure Boot CSP path has a known bug — it exits with error 65000 and silently fails. The policy appears applied in the portal but the UEFI firmware change never happens. Use PowerShell Remediations instead for full control and auditability.

Detection Script

- Checks if Secure Boot is enabled (BIOS/UEFI mode)
- Reads MicrosoftUpdateManagedOptIn registry key
- Reads WindowsUEFICA2023Capable value (stages 3-5)
- Reports current stage (0-5) to Intune portal
- Writes structured log to device for local audit

Remediation Script

- Sets HKLM registry key
MicrosoftUpdateManagedOptIn = 0x5944
- Records timestamp (ManagedOptInDate) for fallback timer
- Verifies write success before reporting exit code
- Handles edge cases: BitLocker, telemetry state
- v3.0 includes direct DB write fallback after threshold

Monitoring & Output

- Compliance shown per device in Intune Remediations
- Structured output: stage, key values, timestamps
- Local log file on device for offline troubleshooting
- BitLocker recovery alerts — keys must be escrowed first
- Track Stage 4 'stuck' devices needing a reboot push

Critical Prerequisites Before You Deploy

1 Enable Required Diagnostic Data (Telemetry)

The managed opt-in (MicrosoftUpdateManagedOptIn) requires telemetry to function. Microsoft's backend uses diagnostic data to detect bad firmware interactions and block unsafe updates. Set AllowTelemetry ≥ 1 (Required). Security level (0) breaks the safety gate. Also ensure the DiagTrack service is running — a disabled service kills the pipeline regardless of policy.

3 Verify BitLocker Recovery Key Escrow

Secure Boot variable changes can trigger BitLocker recovery prompts. Before deploying to any ring, confirm all devices have recovery keys escrowed in Entra ID or AD DS. Document a suspend/resume workflow your helpdesk can execute. Devices entering recovery unexpectedly without accessible keys are a critical risk.

2 Update OEM Firmware First

Firmware limitations and missing OEM-signed KEK entries can block the DB update entirely. Establish an OEM firmware update program with defined rings and validation windows before starting the Secure Boot rollout. Devices on outdated firmware may appear stuck at Stage 2 indefinitely.

4 Cumulative Update $\geq$ February 2024

Windows updates released on or after February 13, 2024 include the capability to install the Windows UEFI CA 2023 certificate. Ensure all devices are current on cumulative updates. In managed environments, the certificate installation is not automatic — the registry opt-in triggers the delivery through Windows Update.

Ring-Based Deployment Strategy

Ring 0 — Pilot



Now

Devices:

IT team + one representative device per hardware model

Actions:

- Deploy remediation script and validate output
- Manually verify each stage transition (0 → 5)
- Confirm BitLocker recovery keys are accessible
- Validate PXE boot and recovery media still work

✓ All Stage 5 with no BitLocker recovery events

Ring 1 — Broad



After Ring 0 validated

Devices:

General employee population (~80% of fleet)

Actions:

- Monitor Intune Remediations dashboard daily
- Alert on devices stuck at Stage 4 (need reboot)
- Watch for unexpected BitLocker recovery prompts
- Set reboot windows aligned with change management

✓ 95%+ compliance within 2 weeks; no critical incidents

Ring 2 — Critical / Sensitive



Final wave

Devices:

High-availability, kiosk, lab, and specialized systems

Actions:

- Require dedicated change window with approvals
- Test dual-boot / custom Secure Boot key devices
- Validate all non-standard boot flows individually
- Plan coordinated reboot to avoid service disruption

✓ 100% compliance with zero unplanned downtime

Watch Out: Common Pitfalls & Edge Cases

Hotpatch & Stage 4 Stall

Windows 11 24H2 with Autopatch Hotpatch delivers security updates without reboots. Devices may sit at Stage 4 (cert staged, reboot needed) for weeks. From the dashboard they look non-compliant — but they're one reboot away. Track Stage 4 separately and push a reboot via Intune.

36-Day Delayed Staging (Real World)

Real-world observation: after setting MicrosoftUpdateManagedOptIn, a device ran 36 consecutive days with Windows Update active — scanning multiple times daily, installing cumulative updates — and never received the Secure Boot certificate payload. The update staging is controlled by Microsoft's backend and is not instant.

BitLocker Recovery Loops

Secure Boot database changes can alter PCR measurements, triggering BitLocker recovery. Escrow all recovery keys BEFORE deployment. Document your helpdesk recovery procedure. Consider suspending BitLocker during the update window for high-risk devices.

Telemetry = Security Level (Value 0)

Enterprise hardening policies often set telemetry to Security level (0). This breaks the managed opt-in safety gate — Microsoft cannot monitor for bad firmware interactions. You must carve out an exception or the certificate update will not reliably deliver through Windows Update.

Custom / Non-Standard Secure Boot Keys

Devices with custom PK, KEK, or DB entries (kiosks, dev machines, dual-boot Linux) need individual assessment. Automated rollout signals may not apply, and incorrect cert injection can render a device unbootable. Flag and handle these manually in Ring 2.

CSP Bug — Error 65000

The built-in Intune CSP for Secure Boot exits with error 65000 on many devices. The Intune portal shows the policy as applied, but the UEFI change never happens. Do not use CSP profiles for this update — use PowerShell Remediation scripts exclusively.

✓ Your Action Plan — Act Before June 2026

TODAY

- Enable Required Diagnostic Data (telemetry ≥ 1) across your fleet
- Confirm DiagTrack service is running on all managed devices
- Escrow all BitLocker recovery keys to Entra ID or AD DS
- Take inventory: OEM models, firmware versions, Secure Boot state

THIS WEEK

- Deploy Intune Remediation scripts to Ring 0 (IT devices)
- Kick off OEM firmware update program for at-risk models
- Review Microsoft Secure Boot Playbook & KB5074109 release notes
- Flag non-standard devices (custom keys, dual-boot, air-gapped)

THIS MONTH

- Validate Ring 0: confirm Stage 5 compliance, no BitLocker events
- Roll out to Ring 1 (general population) with monitoring dashboards
- Set up Stage 4 alert — devices awaiting reboot need a nudge
- Plan Ring 2 (critical systems) change windows with approvals

BEFORE JUNE 2026

- All devices at Stage 5 (WindowsUEFICA2023Capable = in DB, rebooted)
- Server environments separately validated and updated
- Incident response plan ready for any post-deadline edge cases
- Document and close the project — don't let June be a surprise